



Separating compounds

It takes a great deal of effort to separate a compound into pure elements. To make pure iron, you have to separate the compound iron oxide, into iron and oxygen. This is done in a very hot blast furnace.



Iron ore



Pure iron

The mineral iron ore is rich in iron oxide. It is mined to extract iron, which is mostly used to make steel.

Cream and cheese are made by separating milk.



Separating mixtures

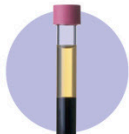
A mixture can be easily separated in several ways.



Evaporation removes water from a mixture by turning it into a gas (water vapor).



Filtration separates large particles, such as coffee grinds, from a suspension.



Spinning at high speed separates blood cells from blood, in a device called a centrifuge.



Distillation separates mixtures of liquids by making them evaporate and condense.



Milk

Strawberries and cream



Separating milk

Whole milk can be separated into cream and skim milk by spinning it in a bowl. The heavier skimmed milk spins away from the lighter cream, which stays in the center of the spinning bowl.